

		images, molecules	
6	(a)	Use the theory of the particulate nature of electromagnetic radiation to explain why there is a threshold frequency for the photoelectric effect.	
			[3]
		Solution: In the particulate theory of light, electromagnetic radiation comprises of discrete quanta of energy, called photons. The energy E of each <i>photon</i> is determined by the frequency f of the radiation, $E=hf$, where h is the Planck's constant. When these photons are incident on the metal surface, they interact with the electrons in the metal on a one-to-one basis, meaning that a photon is either completely absorbed or rejected. (i.e. the energy of different photons cannot be cumulated). For an electron to be ejected from the metal surface, the electron must gain enough energy to be completely freed from the electrostatic forces of attraction of the metal cations. The minimum amount of energy required is called the <i>work function energy</i> Φ that is unique to the type of metal. The value hf must be at equal or larger than Φ. Thus there is a <i>minimum frequency</i> $f=\Phi/h$ the incoming electromagnetic radiation must be at for photoelectric emission to occur.	1 1 1
	(b)	An experimental setup to investigate the photoelectric effect is shown in Fig. 6.1. Electromagnetic radiation is incident on the emitter E of the photocell and the photoelectrons move towards the collector C.	

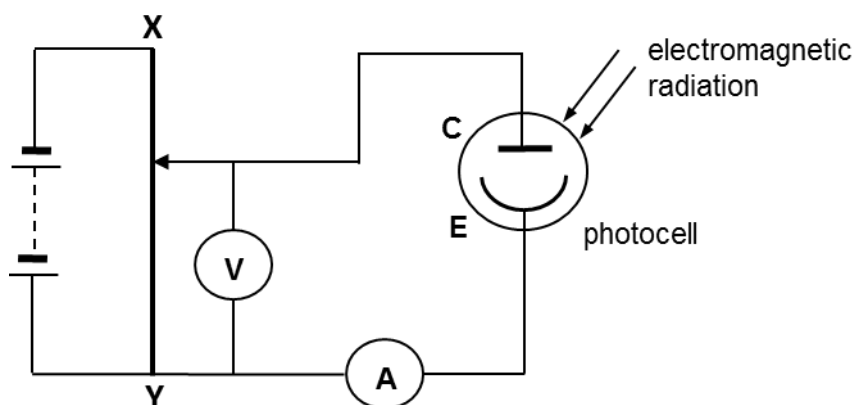


Fig. 6.1

The wavelength of the electromagnetic radiation incident on the photocell was varied. For two values of wavelength λ , the stopping potential V_s required for the ammeter reading to become zero was measured. The results are shown in Fig. 6.2.

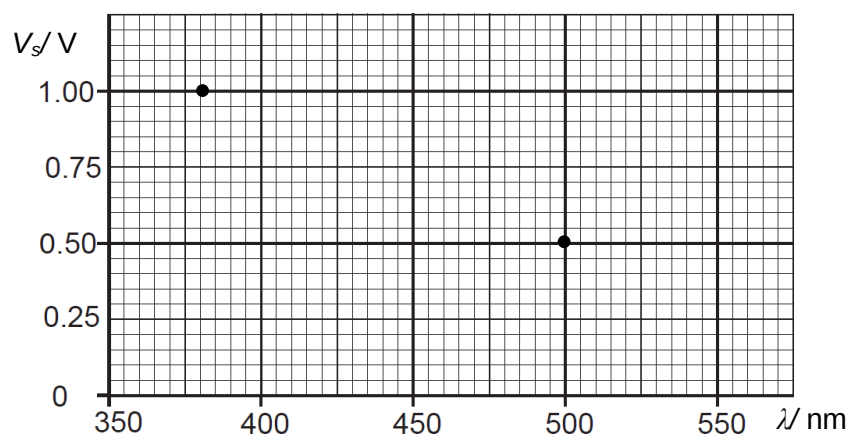


Fig. 6.2

- (i) Calculate the maximum kinetic energy of a photoelectron emitted from the metal surface by radiation of wavelength 500 nm.

maximum kinetic energy = J [2]

Solution:
 Max KE = $eV_s = 1.6 \times 10^{-19} \times 0.5$
 $= 8.0 \times 10^{-20} \text{ J}$

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		(ii)	Calculate the energy of a photon of wavelength 500 nm.	
			energy = J	[2]
			Solution: $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{500 \times 10^{-9}}$ $= 3.98 \times 10^{-19} \text{ J}$	1 1
		(iii)	Hence, determine	
		1.	the work function energy of the metal surface, and	
			work function energy = J	[1]
			Solution: Work function energy = $E_{\text{photon}} - KE_{\text{max}}$ $= 3.98 \times 10^{-19} - 8 \times 10^{-20}$ $= 3.18 \times 10^{-19} \text{ J}$	1
		2.	the maximum photon wavelength that can liberate a photoelectron.	
			maximum wavelength = nm	[2]
			Solution: Work function energy = $\frac{hc}{\lambda_o}$ $\lambda_o = \frac{hc}{\phi} = \frac{6.63 \times 10^{-34} \times (3 \times 10^8)}{3.18 \times 10^{-19}}$ $= 625 \text{ nm}$	1 1

		(iv) Suggest why it is not possible to deduce the maximum photon wavelength that can liberate a photoelectron from this metal plate directly from the data in Fig. 6.2. [1] [Total: 11]	
		Solution: The graph of V_s against λ should give a curve. Since only two points are plotted on Fig 6.2, a curve cannot be drawn to find the value of λ when V_s approaches zero.	1